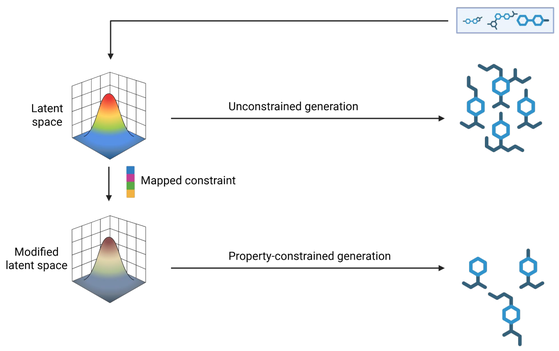
Diffusion-based, property-constrained molecular generation using latent space constraint mapping

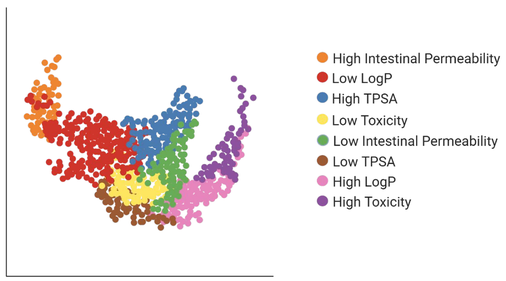
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Task: Map physicochemical & experimental properties into the generative model space



Importance: Enable molecular design from experimental data to increase likelihood of translatable drug generation



Constraining molecular generation to certain regions of the chemical space can result in the preferential generation of molecules that meet experimentally defined objectives

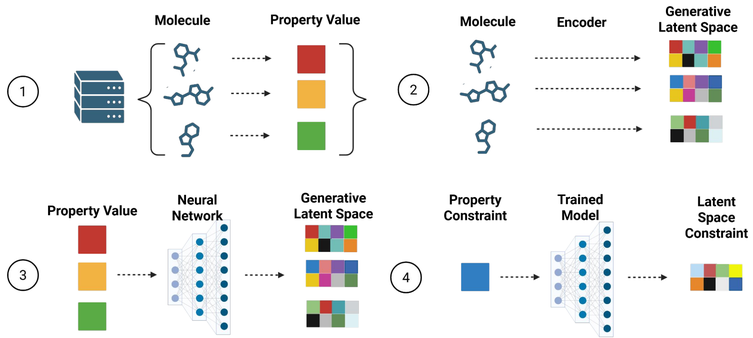
Traditional Molecular Design

Imposing constraints on diffusion models: Applied to image generation but limited by large data requirements not met in domains with scarce experimental molecular property data.

Property-constrained molecular design: Current approaches combine diffusion, equivariant, & LLM models to optimize physicochemical properties with lots of data, but this is hard for experimentally measured traits (ex. toxicity, permeability) due to data scarcity [1, 2].

Conditioning generative model output with a task-specific vector: Current work uses latent space vectors to "guide" output generation [3]. Can be adapted to molecular generation by "mapping" physicochemical & experimental molecular properties into latent space to steer model outputs.

Generating Latent Space Constraint Mapping

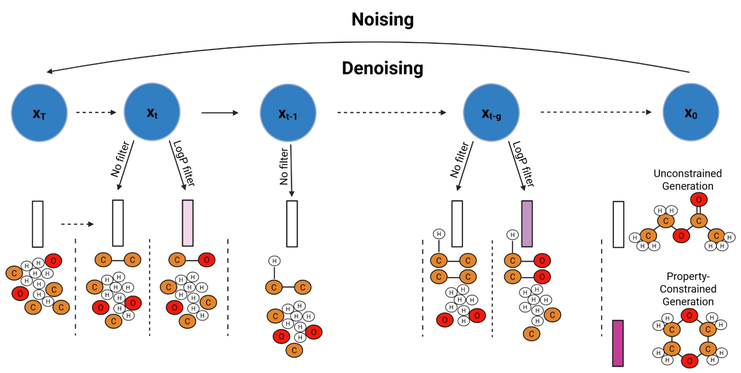


Information from a dataset is used to correlate molecular properties with the latent space representation

Constraint	Dataset	Threshold	Bound Type
Caco-2 Permeability	TDC	-6.0	Lower
Lipophilicity	TDC	2.0	Lower
Aqueous Solubility	TDC	0.0	Lower
VDSS	TDC	5.0	Upper
Acute Toxicity	TDC	2.0	Upper
TPSA	PubChem	50.0	Upper
XLogP	PubChem	0.0	Upper

Various experimentally measured and physicochemical properties are tested to constrain molecular generation

Incorporating Mapping into Molecular Generation for Property Constraint



Mapped latent space constraint matrix is injected into various time steps of the generation process to "condition" the generative process to a certain region of the chemical space

Final Equation for Diffusion Outputs:

$$x'_{t-1} = \begin{cases} (1 - 0.75\exp(\frac{t\log\frac{1}{75}}{1000}))x_{t-1} + \\ 0.75\exp(\frac{t\log\frac{1}{75}}{1000})C_c, & \text{if } t \bmod 5 = 0 \\ x_{t-1}, & \text{otherwise} \end{cases} \qquad x_{t-1} = \frac{1}{\sqrt{\alpha_t}}(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_0(x_t, t)) + \sigma_t z$$

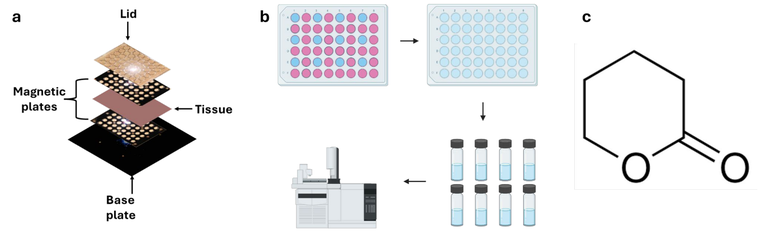
Property-constrained molecular generation results

Property	Type	*LC*	*Original*	$DrugDiff_{pc}$	*DrugDiff*
XLogP	Physicochemical	**0.8224**	0.6927	0.6000	0.0880
TPSA	Physicochemical	**0.8160**	0.2353	0.7640	0.5960
Caco-2 Permeability	Experimental	**0.8477**	0.5050	N/A	N/A
Lipophilicity	Experimental	**0.6907**	0.2982	N/A	N/A
Aqueous Solubility	Experimental	**0.8914**	0.3493	N/A	N/A
VDSS	Experimental	**0.6567**	0.1947	N/A	N/A
Acute Toxicity	Experimental	**0.8359**	0.4026	N/A	N/A

Our algorithm (LC) results in comparable or superior performance when compared with:

1) **Unconstrained molecular generation with same diffusion model (Original)**
2) **Property-conditional molecular generation with DrugDiff ($DrugDiff_{pc}$)**
3) **Unconstrained molecular generation with DrugDiff (DrugDiff)**

Experimental Validation with δ-valerolactone



Testing intestinal permeability of molecule generated using latent space constraint mapping and diffusion model (δ-valerolactone) by measuring passage through segment of porcine jejunum.

Intestinal Permeability Values:

- **Measured: 1.67 x 10^{-5} cm/s**
- **Constraint: ≥10^{-6} cm/s**

$$P_{app} = \frac{V}{A \times C_0} \times \frac{\Delta C_R}{\Delta t}$$

Discussion + Next Steps

- Controlled generation can design molecules with user-defined property thresholds via latent space constraint mapping
- Integrating experimental & molecular physicochemical properties may enhance in vivo success of generated molecules
- **Future work:** 1) More experimental validation, 2) Compare with additional conditional generation baselines, 3) Test more constraints + molecules

References

[1] Alex Morehead and Jianlin Cheng. Geometry-complete diffusion for 3d molecule generation and optimization. Communications Chemistry, 7(1):150, 2024

[2] Marie Oestreich, Erinc Merdivan, Michael Lee, Joachim L Schultze, Marie Piraud, and Matthias Becker. Drugdiff: small molecule diffusion model with flexible guidance towards molecular properties. Journal of cheminformatics, 17(1):23, 2025

[3] Sheng Liu, Haotian Ye, Lei Xing, and James Zou. In-context vectors: Making in context learning more effective and controllable through latent space steering. arXiv preprint arXiv:2311.06668, 2023.